

Unit 2 – Chapter 1

Machine Learning Approaches in NLP

A) Classical NLP Tasks

Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables computers to understand, interpret, generate, and respond to human language. Machine Learning (ML) has significantly improved NLP by allowing systems to learn patterns from data rather than relying only on manually written rules.

1. Text Classification

Definition

Text Classification is the process of assigning predefined categories or labels to text documents automatically using machine learning algorithms.

It is one of the most common NLP tasks and is widely used in emails, news articles, customer reviews, and social media posts.

Examples

- Email → Spam / Not Spam
- News → Sports, Politics, Business
- Reviews → Positive / Negative
- Complaint → Urgent / Normal

Working Process

Text Data



Preprocessing



Feature Extraction



Train ML Model



Predict Category

Steps

Step 1: Collect Dataset

Example

Sentence	Category
India won the cricket match.	Sports
Stock market reaches new high.	Business
New government policy announced.	Politics

Step 2: Text Preprocessing

- Convert to lowercase
- Remove punctuation
- Remove stop words
- Tokenization

- Stemming/Lemmatization

Example

Original:

The players are playing cricket.

Processed:

player play cricket

Step 3: Feature Extraction

Machine learning cannot understand text directly.

Convert text into numbers using

- Bag of Words (BoW)
- TF-IDF
- Word Embeddings

Example

Sentence:

I love NLP

Vocabulary

I

Love

NLP

AI

Vector

[1,1,1,0]

Step 4: Model Training

Common Algorithms

- Naïve Bayes
- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- Neural Networks

Step 5: Prediction

Input

India won the football match.

Prediction

Sports

Applications

- News categorization
- Email filtering
- Document organization
- Customer complaint routing
- Question classification

Advantages

- Fast
- Accurate
- Easy automation
- Reduces manual work

Limitations

- Needs labeled data
- Difficult with ambiguous text
- Sensitive to noisy data

2. Sentiment Analysis

Definition

Sentiment Analysis identifies the emotional tone or opinion expressed in a text.

It determines whether the sentiment is

- Positive
- Negative
- Neutral

Example

Sentence

The movie was amazing.

Output

Positive

Sentence

The food was terrible.

Output

Negative

Sentence

The meeting starts at 5 PM.

Output

Neutral

Workflow

Customer Reviews



Preprocessing



Feature Extraction



ML Model



Positive / Negative / Neutral

Machine Learning Algorithms

- Naïve Bayes
- Logistic Regression
- SVM
- Random Forest
- LSTM
- BERT

Applications

- Product reviews
- Social media monitoring
- Brand reputation
- Political opinion analysis
- Customer feedback

Example Dataset

Review	Sentiment
Excellent product	Positive
Worst experience	Negative
Average quality	Neutral

Advantages

- Understands customer opinion
- Helps businesses improve products
- Fast analysis

Limitations

- Sarcasm detection is difficult
- Mixed opinions confuse models
- Language variations

3. Spam Detection

Definition

Spam Detection identifies unwanted, fraudulent, or irrelevant messages.

Example

Spam Email

Congratulations!

You have won ₹10 lakh.

Click here.

Output

Spam

Normal Email

Meeting scheduled tomorrow at 10 AM.

Output

Not Spam

Workflow

Emails



Cleaning



Feature Extraction



Classifier



Spam / Ham

(Ham = Genuine email)

Common Features

- Frequency of suspicious words
- Number of links
- Special characters
- Sender information
- Capital letters

Algorithms

- Naïve Bayes
- Decision Tree
- Random Forest
- SVM
- Logistic Regression

Applications

- Email services
- SMS filtering
- Social media
- Fraud detection

Advantages

- Protects users
- Saves storage
- Prevents phishing attacks

Limitations

- False positives
- Constantly changing spam techniques

4. Topic Modeling (LDA)

Definition

Topic Modeling automatically discovers hidden topics from a collection of documents without labeled data.

The most popular algorithm is

Latent Dirichlet Allocation (LDA).

It is an **unsupervised learning** technique.

Example

Documents

Football match

Cricket team

World Cup

Topic

Sports

Another set

Bank

Investment

Money

Loan

Topic

Finance

How LDA Works

LDA assumes

- A document contains multiple topics.
- Each topic contains multiple words.

Example

Document

The cricket team won the world cup.

Topic probabilities

Sports → 90%

Politics → 5%

Business → 5%

Workflow

Documents



Preprocessing



Create Document-Term Matrix



LDA Algorithm



Topics Generated

Applications

- News grouping
- Research paper analysis
- Customer feedback
- Legal documents
- Healthcare records

Advantages

- No labeled data needed
- Finds hidden patterns
- Organizes large datasets

Limitations

- Choosing number of topics is difficult
- Topics may be difficult to interpret

5. Keyword Extraction

Definition

Keyword Extraction identifies the most important words or phrases from a document.

Example

Document

Artificial Intelligence is transforming healthcare through machine learning and NLP.

Keywords

- Artificial Intelligence
- Healthcare
- Machine Learning
- NLP

Methods

1. TF-IDF

Selects words with high importance.

2. RAKE

(Rapid Automatic Keyword Extraction)

Finds important phrases based on word frequency.

3. TextRank

Uses graph-based ranking similar to PageRank.

4. Machine Learning

Uses trained classifiers to identify keywords.

Workflow

Document



Cleaning



Tokenization



Word Scoring



Top Keywords

Applications

- Search engines
- SEO
- Document summarization
- Research papers
- Content recommendation

Advantages

- Saves reading time
- Improves searching
- Better indexing

Limitations

- May miss meaningful phrases
- Depends on preprocessing quality

6. Named Entity Recognition (NER)

Definition

Named Entity Recognition identifies and classifies important entities in text.

Entities include

- Person
- Organization
- Location
- Date
- Time
- Money
- Percentage

Example

Sentence

Rohit Sharma plays for India.

NER Output

Word	Entity
Rohit Sharma	Person
India	Location/Country

Another Example

Sentence

Google opened a new office in Mumbai on Monday.

NER Output

Word	Entity
Google	Organization
Mumbai	Location
Monday	Date

Workflow

Sentence



Preprocessing



Feature Extraction



NER Model



Entity Labels

Algorithms

Classical ML

- Hidden Markov Model (HMM)
- Conditional Random Fields (CRF)
- Maximum Entropy Models

Deep Learning

- BiLSTM
- BiLSTM-CRF
- Transformer Models
- BERT
- RoBERTa

Applications

- Search engines
- Chatbots
- Resume parsing
- Medical records
- Question answering
- Information extraction

Advantages

- Automatically extracts useful information
- Reduces manual effort
- Improves search accuracy

Limitations

- Ambiguous names
- Domain-specific entities require custom training
- Multiple meanings of words

Comparison of Classical NLP Tasks

Task	Learning Type	Input	Output	Common Algorithms	Applications
Text Classification	Supervised	Documents	Categories	Naïve Bayes, SVM, Logistic Regression	News classification, Email sorting
Sentiment Analysis	Supervised	Reviews, Tweets	Positive, Negative, Neutral	Naïve Bayes, SVM, BERT	Product reviews, Social media
Spam Detection	Supervised	Emails, SMS	Spam / Ham	Naïve Bayes, Random Forest	Email filtering
Topic Modeling (LDA)	Unsupervised	Documents	Hidden Topics	LDA	Research analysis, News grouping
Keyword Extraction	Unsupervised / Statistical / ML	Documents	Important Keywords	TF-IDF, RAKE, TextRank	SEO, Search engines
Named Entity	Supervised	Sentences	Named Entities	CRF, BiLSTM,	Chatbots, Search

Recognition (NER)				BERT	engines, Information extraction
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